



one type-1 individual, horizon $t + \Delta t$
resolve the *first* event in the interval $[0, \Delta t]$

$$1 - (\lambda_1 + \mu_1 + \nu + \delta_1)\Delta t$$

$$\lambda_1 \Delta t$$

$$\mu_1 \Delta t$$

$$\nu \Delta t$$

$$\delta_1 \Delta t$$

Nothing



still one type-1

contributes
 $S(t)$

Birth



two independent
type-1 lineages

contributes
 $S(t)^2$

Death



population empty;
catastrophe impossible

contributes
 1

Conversion



type-1 lineage
becomes type-2

contributes
 $G(t)$

Catastrophe



absorbed; the event
already failed

contributes
 0

$$S(t + \Delta t) = \underbrace{[1 - (\lambda_1 + \mu_1 + \nu + \delta_1)\Delta t]}_{\text{nothing}} S(t) + \underbrace{\lambda_1 \Delta t}_{\text{birth}} S(t)^2 + \underbrace{\mu_1 \Delta t}_{\text{death}} \cdot 1 + \underbrace{\nu \Delta t}_{\text{conversion}} G(t) + \underbrace{\delta_1 \Delta t}_{\text{catastrophe}} \cdot 0 + o(\Delta t)$$



subtract $S(t)$, divide by Δt , let $\Delta t \downarrow 0$

$$\frac{dS}{dt} = \lambda_1 S^2 - (\lambda_1 + \mu_1 + \nu + \delta_1)S + \mu_1 + \nu G$$